SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Claypot Chicken Rice
Project / Product Name	Jobee
Final Live Deployment URL	https://jobee-sdchack.replit.app/
Codebase Repository URL	https://github.com/jimmerino2/sdc_hackathon_2026
Name (Leader)	Lam Way Hou
Email & phone (Leader)	lamwayhou@gmail.com +60162211236

2. Project Description

[Maximum 250 words]

In today's world, AI is changing and adapting constantly, changing which tech jobs are hiring or shrinking, at a pace where courses and career advice can't keep up. This leaves students and fresh graduates under-skilled for positions today. This issue is common, especially in the tech industry, including but not limited to software developers, data scientists, and other positions. To combat this, Jobee was built.

Jobee starts by onboarding users and profiling them based on their interests, skills, and educational background. Finally, it matches them against tech roles. Each role is scored by several metrics such as the skills you have, interests tied to the 17 SDGs, as well as whether the job itself is rising or falling under the influence of AI. A forecasting pipeline built on published AI-adoption and job-postings data supplies the regression models, providing evidence-backed forecasting that ensures that the jobs will be future-proof.

Jobee then provides a personalized, tailored roadmap to help users reskill themselves to the role. This includes providing verified and effective courses, videos, and other sources to help learn each skill for the job. Each section opens with an AI skill, teaching users to learn faster using AI tools. To validate that the users are reskilled, special challenges that require users to create customized, SDG and interest-tailored projects are done. Through gamification and encouragement of multiplayer and SDLC principles, we believe Jobee is the right tool for our users to reskill, upgrade, and seamlessly enter the industry regardless of AI influence.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Jobee is a web application with three connected layers: a Vue 3 frontend, a FastAPI backend and a Supabase database. Around them sit a GenAI gateway and an offline data pipeline.

The frontend is a single-page app built with Vue 3, Vite and TypeScript. It guides a learner from profile and resume upload through ranked roles, a market forecast and a skill-gap check to a personalized learning pathway. Long tasks, such as reading a resume or building a roadmap, stream their progress to the screen live.

The backend is a FastAPI service organized into modules for resumes, role matching, forecasting, roadmaps, challenges, HRD Corp programmes, SDG mapping and LinkedIn sharing. Role ranking is plain arithmetic, combining skills, job-posting outlook, interests and education, so it is repeatable and explainable.

Supabase provides the Postgres database and authentication. It stores profiles, roadmaps, role and skill reference data, SDG weights, HRD Corp programmes and stored forecast signals. Learners sign in with an emailed one-time code, so there are no passwords.

GenAI is used in five places, all through one OpenAI-compatible gateway to Gemini 3.1 Flash-Lite. It reads uploaded resumes and suggests skills, arranges skills into a roadmap with sections and prerequisites, writes an AI skill for each roadmap section, writes and judges starter projects, and explains why a role suits an interest. It never decides the ranking.

AI-assisted learning puts one AI skill first in each roadmap section, teaching learners to use AI tools to learn that section's skills, and the section's other skills stay locked until it is complete. AI skills are stored by role for reuse. Their source links come from ESCO, the European Commission's public skills taxonomy, never from the model. If the model or ESCO is unavailable, the roadmap still builds.

Forecasting follows a two-step design. Offline pipeline scripts turn Indeed Hiring Lab postings and AI-tracker data into sector forecasts and join them to O*NET jobs. At runtime, the API reads these precomputed tables. On the Signal page, a LightGBM model retrains in memory, in about two seconds, on whichever quarters the learner keeps. The three AI futures are labelled assumptions, not model output.

Finally, the app integrates the public HRD Corp programme list and optional LinkedIn sharing through OAuth, which posts once and stores no token. Privacy is handled by a consent checkbox and a Privacy Policy page, and resume files are never saved.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

We opened the finalist sprint by auditing our Phase 1 build against the finalist guide's must-haves, then closed the gaps it found.

Personalized, time-bound learning. Roadmaps now adapt to the learner: skills are suggested from their profile and editable before generation, each skill shows a time estimate, and recommended skills are weighted by O*NET importance and demand trends. A what-if simulator and match-score explanations show why each role ranks where it does.

AI-assisted learning. Each roadmap section now opens with an AI skill that teaches learners to use AI tools to learn that section's skills, with source links from ESCO. The simulator shows how they improve role fit, and round 2 challenges now require an AI feature.

Trends that visibly change. The Signal page lets users include or exclude years and quarters, and the LightGBM forecast retrains live on that selection in about two seconds. A sync button loads test data so the effect of changed data can be shown on demand.

Affordable, SDG-linked pathways. Free resources are labelled, and HRD Corp Verified programmes appear beside the skills they teach. Roles and skills map to SDG sub-targets, and a role mind-map shows how roles connect.

Sharing and progress. Learners can share a completed roadmap and challenges to LinkedIn after consenting. Completed skills stay viewable, and challenges can be filtered by current or past.

Pivots from Phase 1. After our RunPod credits ran out, we removed that dependency and moved every AI call to one gateway model (Gemini 3.1 Flash-Lite), so each plan fits its learner. We simplified the interface, navigation and tabs after team testing.

Responsible by design. We added a PDPA consent checkbox and a Privacy Policy page and costed running at about $0.02–0.04 per learner.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Baseline taken from Phase 1: Skill demand prediction accuracy: 77.5% AI adoption prediction accuracy: 85% AI acceleration prediction: 74.1% (3 months), 68.9% (6 months), 54.5% (12 months)
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	**AI capability** (METR task time horizon, 3, 6 and 12 months ahead) • **Log:** frontier_time_horizon/ • **Proves:** the ensemble beat the naive forecast on the holdout, with 45–71% lower error. The sample is small: 6 to 16 scored points per horizon. **Business AI adoption** (US Census BTOS, 6 and 13 fortnights ahead) • **Log:** btos_ai_use_rate/ • **Proves:** the forecasts were tested honestly, and nothing beat naive. At 13 fortnights the trend model's error was 1.70× naive, so naive is what's served. **Worker AI adoption** (US generative AI use by occupation group, 1 quarter ahead)

Metric Category	Your Response
	• **Log:** worker_genai_use_rate/ • **Proves:** the same as business adoption. pooled_mean narrowly lost to naive, so naive is served. With only 7 origins, this is a weak test. **Job postings forecast** (LightGBM, Indeed index by sector, 3, 6, 9 and 12 months ahead) • **Log:** model_meta.json, with the scores explained in ml/README.md. • **Proves:** LightGBM was scored on a 12-month holdout against three baselines, and all four horizons are marked indicative, not validated. And for all the forecasting result is forecasted via multiple methods with naive being the ground truth and all of them undergoes comparison before selection.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	1.Community & Institutional Impact Our platform transforms from an individual tool into a public-good asset by addressing structural economic challenges: • Bridging the Youth Unemployment Gap: By analyzing real-time labor market trends and delivering personalized skill-learning loops, the system helps bridge the disconnect between academic curricula and actual industry demand. • Democratizing Career: Providing AI-driven, elite-level resume parsing and career trajectory mapping grants underrepresented jobseekers access to high-quality career guidance that is traditionally locked behind expensive consulting services. • Building AI Literacy: Every roadmap section opens with an AI skill that teaches learners to use AI tools to learn faster and understand more, not to

Metric Category	Your Response
	do the work for them, so jobseekers without paid training gain practical AI habits as they reskill. 2.Civic & Corporate Partner Ecosystem The API service plugin model enables multi-sector stakeholder support and funding: • Government & Educational Alliances: Public employment agencies, municipal workforce development boards and university career centers can sponsor the deployment of our API plugin across public portals to upskill regional talent pools. • Corporate CSR & ESG Sponsorship: Enterprise partners seeking measurable Corporate Social Responsibility (CSR) impact can fund platform access for underserved communities directly fulfilling their ESG human-capital metrics. 3)Scalable Real-World Deployment Roadmap Designed for rapid horizontal expansion across institutional systems: I. Phase 1: Institutional Integration (API Embed) Deploy the API Plugin directly into target partner ecosystem (e.g. University, Career Services portals, municipal job boards) II. Phase 2: Localized Trend Analytics Ingest regional labor market datasets to tailor skill recommendations specifically to local economic development priorities. III. Phase 3: Ecosystem Scale Establish data-sharing feedback loops with corporate HR partners to create a direct talent pipeline from learning pathways into hiring funnels.

Metric Category	Your Response
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	a) System Integration & Architecture To maximize adoption, our solution is engineered as a lightweight, modular API Service Plugin, rather than forcing organizations to replace their existing infrastructure, our tool acts as an interoperable middleware layer that integrates smoothly into current environments. Plug-and-Play Integration which easily connects with major IT Asset Management (ITAM) platforms, enterprise resource planning (ERP) tools and internal corporate portals via standard RESTful endpoints. Stateless and Scalable built with Fast API to ensure high throughput, asynchronous request handling and automatic OpenAPI documentation for developer-friendly onboarding. b) AI-Driven Resume Parsing & Positioning AI-Driven Deep Resume Analysis powered by advanced AI algorithmic model, the system performs semantic-level parsing on user resumes and accurately identifying core skills and experience gaps to provide data-backend career positioning. c) Trend Forecasting & Skill Development Career Trend Prediction & Skill Learning Loop by analyzing macro-market trends and individual interest profiles, the AI algorithm predicts future career trajectories and delivers tailored skill learning recommendations which helping talent proactively adapt to a rapidly evolving labor market. d)B2B / B2B2C Commercialization Operating as an embedded plugin, the system offers corporate HR teams an efficient AI-powered talent screening tools or providing university career centers with student competitiveness analytics by unlocking vast SaaS commercialization prospects and high ROI

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

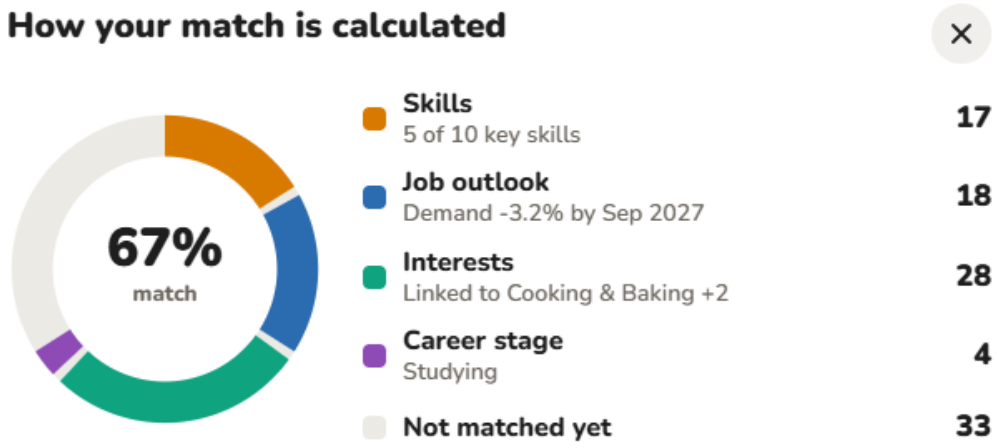
This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Job Matching Score:

score = (0.10 * education + 0.25 * interests + 0.35 * skills + 0.35 * outlook) / 1.05 (each part 0-1)

Details:

1. All weights and set values are in **src/backend/app/jobs/scoring.py.**
2. Education weights: student 0.4, fresh graduate 0.6, working 1.0 (the stage level divided by 2.5).
3. Skills proficiency rates: Basic 0.33, Good 0.67, Strong 1, none 0, set by the user.
4. Skills: Σ(importance × proficiency) ÷ Σ importance over the role's 10 key skills. Importance is the O*NET rating, doubled if demand is forecast to rise.
5. Outlook: the Indeed postings forecast for the chosen AI future, converted to 0–1. Flat postings score 0.5.
6. Interests: how closely the learner's interests match the SDGs the role serves, compared with the best-matching role.
7. Sample calculation:



Job Skills Decision-making:

1. Currently, only 8 roles are selected after importing O-Net Database, because we decided to pivot and focus on developers, below is the list of roles + their ONet code

src\backend\app\jobs\service.py

```python
DATA_DIR = Path(__file__).resolve().parents[4] / "data" / "processed"
TOP_JOBS = 10
DEFAULT_ICON = "briefcase"
KEY_TOOLS = 6  # a role's key skills: this many tools ...
KEY_ABILITIES = 4  # ... plus this many abilities (O*NET skill_type "skill", e.g. Programming, Critical Thinking)
# Only tech roles are matched for now.
TECH_ROLES = {
    "15-1211.00",  # Computer Systems Analysts
    "15-1212.00",  # Information Security Analysts
    "15-1232.00",  # Computer User Support Specialists
    "15-1252.00",  # Software Developers
    "15-1255.00",  # Web and Digital Interface Designers
    "15-2031.00",  # Operations Research Analysts
    "15-2051.00",  # Data Scientists
    "15-2051.01",  # Business Intelligence Analysts
}
```

2. This also indirectly limits what skills we recommend, as skills are tied to roles
3. We then filter for skills that ONET marked importance > 2.25 to reduce what we send to the LLM, and filter off redundant skills
4. Each skill carries a forecast-demand value, from Hacker News job-ad trends, that is given to the LLM as a hint. (Separately, the match score doubles the weight of rising skills. That is not part of roadmap selection.)

```python
@lru_cache
def is_rising(onet_code: str, skill_name: str) -> bool:
    """Whether the Occupations page forecasts rising demand for this skill in this role's occupation (evidence above low)."""
    occupation = OCCUPATIONS.get(onet_code)
    if occupation is None:
        return False
    rows = [r for r in _demand().get(REGISTRY_NAMES.get(skill_name.lower(), skill_name.lower()), []) if r["occupation"] == occupation and r["evidence"] != "low"]
    return bool(rows) and rows[0]["direction"] == "rising"      You, 10 hours ago • fix: Skills recommended now relative to ONET ro…
```

5. For any skills with proficiency > 0.67, we take it that the user already knows said skill, and excludes it when generating a new roadmap
6. Finally, the LLM decides which skills to include, then groups the skills by segments and builds the roadmap

```python

def suggest_skills(db: Client, profile_id: str, onet_code: str) -> SkillSelection:
    """The skills to include in this learner's roadmap for a role, as a plain list (no sections, no order).

    The model picks them from the role's O*NET skills, seeing what the learner already knows; skills they know well are
    left out. If the model can't be used, the role's most important skills they lack are taken instead (source "fallback").
    """
    try:
        profile = profiles.get_profile(db, profile_id)
    except profiles.ProfileError as e:
        raise HTTPException(e.status_code, e.message) from e
    role = _role(db, onet_code)
    rows = db.table("onet_role_skills").select("skill_id,skill_name,skill_type,importance,level,forecast_demand").eq("onet_code", onet_code).execute().data
    relevant = planner.candidates(rows)
    if not relevant:
        raise HTTPException(status_code=422, detail=f"no skills above importance {planner.MIN_IMPORTANCE} for {onet_code}")
    levels = learner_levels(profile)
    have = {s["skill_id"]: levels[s["skill_name"].lower()] for s in relevant if levels.get(s["skill_name"].lower())}
    try:
        ids, source = planner.select_skills(role["title"], role["description"], relevant, {"stage": profile.stage, "have": have}), "ai"
    except HTTPException as e:
        print(f"skill selection for {onet_code} fell back ({e.detail})")
        ids, source = planner.fallback_selection(relevant, have), "fallback"
    ids = [i for i in ids if have.get(i, 0) < planner.OWNED_LEVEL]  # the model was told, but this is the rule
    by_id = {s["skill_id"]: s for s in relevant}
    return SkillSelection(
        onet_code=onet_code, title=role["title"], source=source,
        skills=[SelectedSkill(skill_id=i, skill_name=by_id[i]["skill_name"], skill_type=by_id[i]["skill_type"]) for i in ids],
    )
```

AI Capability and Acceleration Forecast:

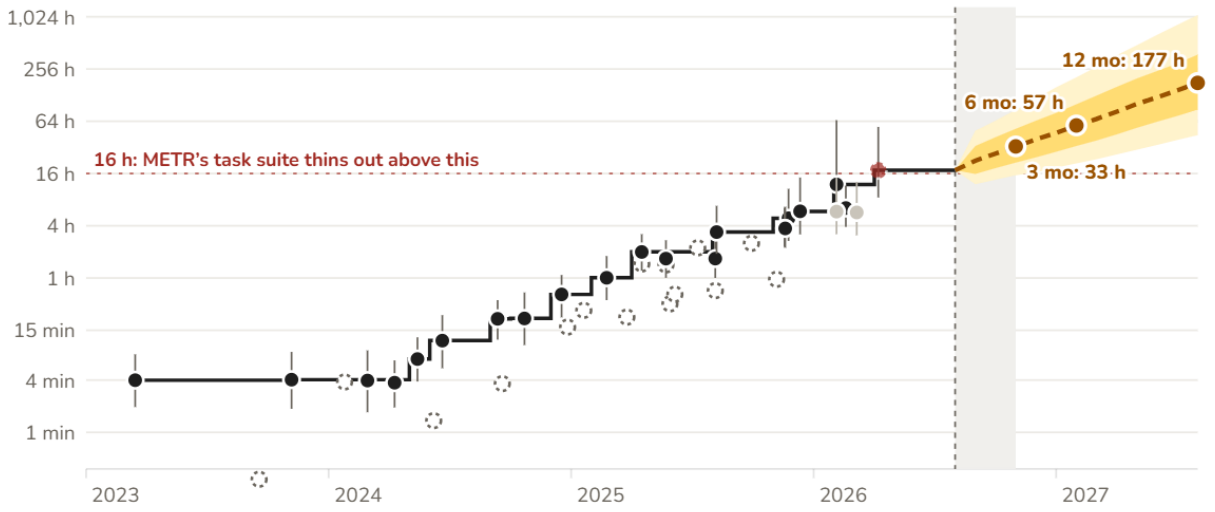
AI capability refers to "the longest task the best AI model can finish with a 50% success rate", measured by METR software task.

all the data received are split into:

- training data
- test data
- final exam data

After normalizing the data, it is used to build a frontier where it represents the black stripes of the chart below, which is also the best score so far, held flat until a model beats it. That makes the staircase line chart: flat for months, then a jump. Meanwhile, these data are used to train the 12 chosen forecasting methods and tested with the test data to select the 1 single best forecasting method to forecast the predictions.

```python
def frontier(rows: pd.DataFrame, last: pd.Period) -> pd.DataFrame:
    """Monthly frontier (max value among models released by month end), first release month .. last."""
    if rows.empty:
        return pd.DataFrame(columns=["period", "value"])
    rel = rows.assign(period=rows["release_date"].dt.to_period("M"))      Li Yang, last week • feat(forecasting): AI ca
    months = pd.period_range(rel["period"].min(), last, freq="M")
    by_month = rel.groupby("period")["value"].max().reindex(months)
    best = by_month.ffill().cummax()  # no release: frontier holds
    # ... but only while a measured model is recent. METR measured almost nothing from 2019-2022, so before GPT-4
    # the "frontier" would be GPT-2 (2019); a later jump from it says nothing about the pace of progress.
    last_release = pd.Series(months, index=months).where(by_month.notna()).ffill()
    stale = [(m - r).n > MAX_STALE_MONTHS if pd.notna(r) else True for m, r in zip(months, last_release)]
    best[stale] = float("nan")
    return pd.DataFrame({"period": months, "value": best.to_numpy()})
```



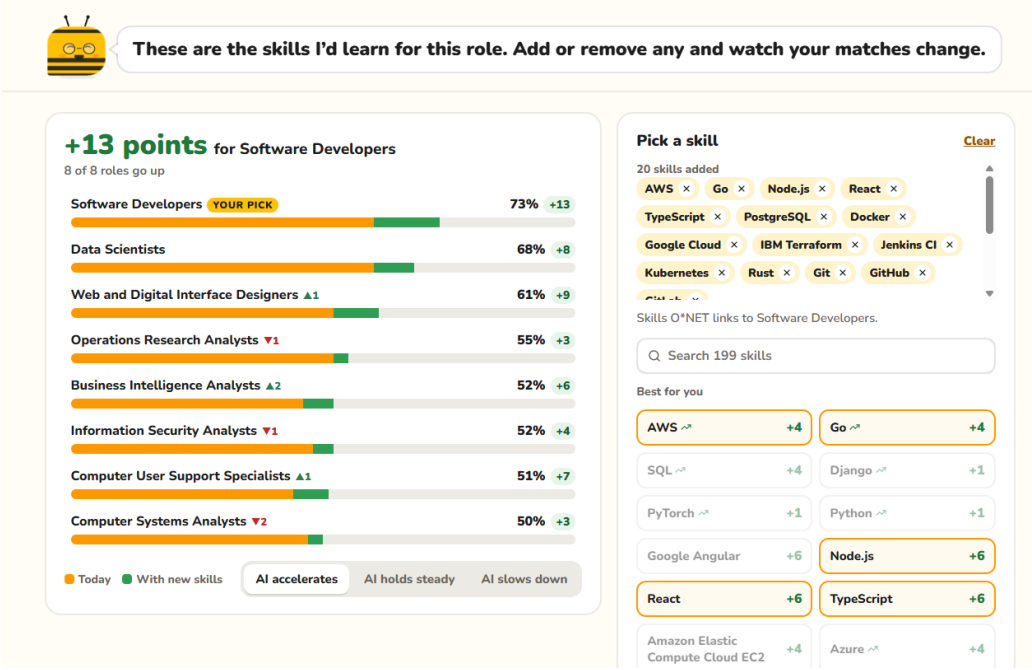
The selected forecasting method then undergoes 5000 shares of Monte Carlo Simulations with the variable factor of:

- Starting point on the simulation
- Future growth factors

And throughout the 5000 simulations, it would take the current prediction value, the +20% and –25% range would be as the same pace, and over or beneath it would indicate AI is speeding up or slowing down.

Skills Affecting Jobs Score:

1. This is referring to this page for context:



2. Each key skill's weight is its O*NET importance. Tools rated "In Demand" are 4.0 and "Hot Technology" tools are 3.5.
3. If the skill demand is forecast to rise, which comes from Hacker News job ads, its weight is doubled.

src\backend\app\jobs\service.py

```
    # Trend momentum: a skill forecast to be in rising demand for this role counts double, so learning it moves the match more.
    trending = {n for n, _ in picked if is_rising(role["onet_code"], n)}
    return {**role, "key_skills": [n for n, _ in picked], "trending": trending,
            "weights": {n: w * (TREND_WEIGHT if n in trending else 1.0) for n, w in picked}}
```

You, last week • feat: Hooking up the UI from Replit to the back…

4. The weights are then converted into shares via this formula share = weight / Σ weights of the 10 skills
5. Finally, the user's proficiency in the score is evaluated with this formula skills score = Σ (weight × proficiency) / Σ weight
6. For example, Python is In Demand (4.0) and rising, so its weight is 4 × 2 = 8. Dividing the total weight keeps the score between 0 and 100% so the sum does not overflow > 100%

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Jobee currently uses US-job posting data to substitute for Malaysia data and only limited 8 tech and IT related roles. Fundings and partnerships would go first to replacing that because local data adds more value, reliability and trust than extra infrastructure.

The data source is also highly insufficient; we would build a scheduled pipeline that collects jobs ads continuously, extract skills with an LLM, and maps them to one skill vocabulary. Time passes and we could have more data to back up our predictions and recommendations.

Because forecasts are precomputed, learner traffic never triggers a heavy batch of work or latency issues. We estimate around 0.02-0.04 costing per learner per month, which has tremendously low operational costing, which facilitates faster scalability.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- [X] The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- [X] The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- [X] The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- [X] The final pitch deck has been exported and saved as a flattened PDF file.
- [X] All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Indeed Job Postings Index (Indeed Hiring Lab)	dataset	CC-BY-4.0
Indeed AI Tracker (Indeed Hiring Lab)	dataset	CC-BY-4.0
O*NET 31.0 Database (USDOL/ETA)	dataset	CC-BY-4.0
Anthropic Economic Index	dataset	CC-BY (version to confirm)
US Census Business Trends and Outlook Survey	dataset	Public domain (US government work)
Real-Time Population Survey on generative AI	dataset	-
METR time-horizon results	dataset	-
DOSM Job Market Insights (Malaysia)	dataset	-
Stack Overflow Developer Survey 2023–2025	dataset	ODbL
Hacker News "Who is hiring?" threads (via Algolia API)	dataset	Public API

HRD Corp BizMatch public programme list	dataset	All rights reserved (used as a reference, with links to the source)
DeepSeek V4.1 Flash	model (API)	KrackedDevs
LightGBM forecasting model	model (trained by us with the datasets above)	Our own weights; library license below
Vue 3, Vue Router, Vite, @vitejs/plugin-vue, vue-tsc, Vitest	code	MIT
TypeScript	code	Apache-2.0
lucide-vue-next (icons)	code	ISC
pdfjs-dist (PDF.js)	code	Apache-2.0
FastAPI, Pydantic, pydantic-settings, openpyxl, supabase-py	code	MIT
Uvicorn (via fastapi[standard])	code	BSD-3-Clause
pandas, NumPy, scikit-learn, pypdf	code	BSD-3-Clause
LightGBM	code	MIT
OpenAI Python SDK	code	Apache-2.0

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



01/11/2026